

Aegis Grid: Software Guide

1. The Sensor Grid (8 LDRs)

Logic: A 4 x 4 grid is mapped using 4 sensors for the X-axis (Columns) and 4 for the Y-axis (Rows).

- **Wiring per LDR:**
 - One leg to **5V**.
 - Other leg to **Input Pin** of Encoder **AND** a **10k Ω Pull-down resistor** to Ground.
- **Grid Mapping:**
 - **Column Sensors (X):** Inputs for **Encoder 1**.
 - **Row Sensors (Y):** Inputs for **Encoder 2**.

2. The Priority Encoders (2x 74LS148)

Logic: These convert the active sensor into a 2-bit binary coordinate. Note: 74LS148 is **Active-Low** (it triggers on 0V).

Pin Mapping (Same for both X and Y Encoders):

- **Pin 16 (VCC):** 5V (Input)
- **Pin 8 (GND):** Ground (Input)
- **Pin 5 (EI):** **Ground** (Input - Must be grounded to enable the chip)
- **Pins 10, 11, 12, 13 (I0, I1, I2, I3):** Connect to Sensors 0, 1, 2, 3 (Inputs)
- **Pins 1, 2, 3, 4 (I4, I5, I6, I7):** **5V** (Input - Keeps higher priorities disabled)
- **Pin 9 (A0):** Binary Bit 0 (Output)
- **Pin 7 (A1):** Binary Bit 1 (Output)
- **Pins 6, 14, 15:** Leave Disconnected.

3. The Bit Inversion (74LS04 NOT Gate)

Logic: Because the 74148 is active-low, its outputs are inverted (e.g., Sensor 3 = **00**). We use NOT gates to return them to "Normal Binary" (Sensor 3 = **11**).

- **Connections:**
 - **A0 and A1 (X-Encoder)** Through NOT gates **B0 and B1** of Comparator.
 - **A0 and A1 (Y-Encoder)** Through NOT gates **B2 and B3** of Comparator.

4. The "Brain" (74LS85 4-Bit Comparator)

Logic: Compares the Target Position (B) from sensors against the Guard Position (A) from switches.

Pin Mapping:

- **Pin 16 (VCC):** 5V (Input)
- **Pin 8 (GND):** Ground (Input)
- **Input A (DIP Switches):** Pins 10, 12, 13, 15 (Connect to 4 switches with pull-down resistors).
- **Input B (Sensor Data):** Pins 9, 11, 14, 1 (Connect to the outputs from the NOT gates).
- **Cascade Inputs (CRITICAL for Software):**
 - **Pin 3 (A=B IN):** 5V (Input)
 - **Pin 2 (A<B IN):** Ground (Input)
 - **Pin 4 (A>B IN):** Ground (Input)
- **Outputs (Status LEDs):**
 - **Pin 5 (A>B OUT):** To Yellow LED (Output)
 - **Pin 6 (A=B OUT):** To Green LED & AND Gate (Output)
 - **Pin 7 (A<B OUT):** To Blue LED (Output)

5. The Fire Logic (74LS08 AND Gate)

Logic: The system only "Fires" if the target is **Locked** AND the **Safety Switch** is ON.

- **Pin 1 (Input 1):** From **Pin 6 (A=B)** of the Comparator.
 - **Pin 2 (Input 2):** From a **"Master Arm"** Toggle Switch.
 - **Pin 3 (Output):** To Buzzer or Red "Fire" LED.
-